

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10: REAL NUMBERS FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished the unit on real numbers. Now answer the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important? Use Achieng's trip to the duka near Kibuye Market in Kisumu as your main example. Write your answer in the space under each prompt. Use evidence and numbers from your lessons, your Driving Question Board (DQB) and your unit model. Show your working where you calculate. Write clearly and explain your reasoning in full sentences.

Part 1: Different numbers at the duka

Achieng bought 3 items at Ksh 45 each, paid with one Ksh 200 note, bought $\frac{1}{2}$ kg of sugar at Ksh 160 per kg, and saw a tea-leaf sachet tagged Ksh 22.50. (a) Work out her total for the 3 items and her change. (b) Name the type of number used for counting items, for part of a kilo, and for the price tag, and explain why one trip needs all three types. (c) Explain how a fraction and a decimal can name the same quantity, using one example from the trip.

(a) Total = $3 \times 45 = \text{Ksh } 135$. Change = $200 - 135 = \text{Ksh } 65$. (b) Whole numbers are used for counting items and money amounts such as 3, 135 and 65. A fraction, $\frac{1}{2}$, is used for a part of a whole, half a kilo of sugar, which costs $\frac{1}{2} \times 160 = \text{Ksh } 80$. A decimal, Ksh 22.50, is used for a price with cents. One trip needs all three because counting, sharing a whole into parts and pricing with cents are different jobs. (c) They are two ways of writing the same rational number. For example $\frac{1}{2} = 0.5$, and $\text{Ksh } 22.50 = 22 \frac{1}{2} = 45/2$. All of these numbers are real numbers.)

Part 2: Classifying whole numbers

Some whole numbers from Achieng's trip are 3, 6, 15, 45, 65, 90, 135 and 200. (a) Classify 3, 15 and 90 as odd or even, and as prime or composite, and justify each using factors. (b) Explain why 1 is neither prime

(a) 3 is odd and prime because its only factors are 1 and 3. 15 is odd and composite because its factors are 1, 3, 5 and 15 ($15 = 3 \times 5$). 90 is even and composite because it is divisible by 2 and has many factors (for example 1, 2, 3, 5, 6, 9, 10...). (b) 1 has only one factor, itself, so it does not have exactly two factors (prime) or more than two factors (composite). 2 is the only even prime because its factors are 1 and 2, while every other even number is divisible by 2 and so has at least three factors. (c) Seventeen cobs cannot be shared equally among 2 to 16 groups without a remainder. This shows 17 has only the factors 1 and 17, so it is prime. (d) No. Being odd does not make a number prime. The odd duka numbers 15, 45, 65 and 135 are composite: $15 = 3 \times 5$, $45 = 3 \times 3 \times 5$, $65 = 5 \times 13$, and 135 has more than two factors. Only 3 among them is prime.)

nor composite, and why 2 is special. (c) Seventeen cobs of maize are to be shared equally among groups. What happens, and what does this tell you about 17? (d) Is every odd number prime? Use the duka numbers to support your answer.	
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Part 3: Rational and irrational numbers	
(a) Use division to explain what happens to the decimals of $\frac{1}{4}$ and $\frac{1}{7}$, and say whether each is rational. (b) What is the definition of a rational number? Show that the whole number 3 and Ksh 22.50 are rational. (c) Squaring 1.414, 1.4142 and 1.415 on a calculator gave 1.999396, 1.99996164 and 2.002225. Using these results and the square plot of area 2 m², explain why $\sqrt{2}$ goes on forever and what type of number it is. (d) Describe how rational and irrational numbers together make up the real numbers, and place 3, $\frac{1}{2}$, 22.50, $\sqrt{2}$ and π on this map.	(a) $\frac{1}{4} = 0.25$, which ends (terminating). $\frac{1}{7} = 0.142857142857\dots$, which never ends but repeats the block 142857 (recurring). Both are rational. (b) A rational number can be written as $\frac{a}{b}$ with a and b integers and $b \neq 0$; its decimal terminates or recurs. $3 = \frac{3}{1}$ and $22.50 = \frac{45}{2}$, so both are rational. (c) The side of a square plot of area 2 m² is $\sqrt{2}$ m. 1.4142 squared is just under 2 and 1.415 squared is over 2, so the side lies between 1.4142 and 1.415. No ending decimal squares to exactly 2, and the digits of $\sqrt{2} = 1.41421356\dots$ do not repeat in a block, so it cannot be written as $\frac{a}{b}$. It goes on forever because it is irrational. (d) Rational numbers (whole numbers, fractions, terminating and recurring decimals) and irrational numbers together make up the real numbers. On the map, 3, $\frac{1}{2}$ and 22.50 are rational; $\sqrt{2}$ and π (3.14159265...) are irrational. All are real numbers.)

Part 4: Reciprocals and the tools to find them	
(a) What is the reciprocal of a number n, and what do you get when you multiply n by its reciprocal? Which number has no reciprocal? (b) Find the reciprocal of 8, of 2.5 and of 6 (give 6 to 4 decimal places) and check one of them by multiplying. (c) Mzee Odhiambo says 6 mandazi cost Ksh 90, so	(a) The reciprocal of n is $1 \div n$, written $\frac{1}{n}$, and $n \times \frac{1}{n} = 1$. Zero has no reciprocal. (b) The reciprocal of 8 is 0.125; of 2.5 is 0.4; of 6 is 0.1667 (4 d.p., since $\frac{1}{6} = 0.1666\dots$). Check: $6 \times \frac{1}{6} = 1$ (and $2.5 \times 0.4 = 1$). (c) Unit price is total cost $\div$ number of items: $90 \div 6 = \text{Ksh } 15$. This is the same as multiplying by the reciprocal: $90 \times \frac{1}{6} = \text{Ksh } 15$. (d) Mathematical tables give $\frac{1}{7} \approx 0.1429$ (4 significant figures) and the calculator gives 0.142857142... with more digits. They differ because $\frac{1}{7}$ is a recurring decimal that never ends, so every tool must round it. Both are accurate enough for money, and the calculator shows more digits. We should look after the tables and calculators carefully.)

the unit price is Ksh 15. Explain how this is a reciprocal calculation. (d) Compare the answers for $1/7$ from mathematical tables and from the calculator $1/x$ key. Why do they differ, and are both useful for money?	
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Part 5: Answering the Driving Question	
(a) A bus travels from Nairobi to Mombasa, 480 km. Use time = distance $\times$ $1/\text{speed}$ to find the time at 60 km/h and at 80 km/h. (b) Wanjiru saves Ksh 10 000 in a Sacco at 8% simple interest per year. Find the interest for 1 year and for 6 months, then use a reciprocal to find how long she needs to earn Ksh 2 400. (c) Write a final answer to the Driving Question: How do we use real numbers in everyday life, and why are reciprocals important? Connect it back to Achieng's trip.	(a) At 60 km/h: $480 \times 1/60 = 8$ hours. At 80 km/h: $480 \times 1/80 = 6$ hours. (b) Interest for 1 year = $10\,000 \times 8/100 = \text{Ksh } 800$. For 6 months it is Ksh 400. Time to earn Ksh 2 400 = $2\,400 \times 1/800 = 3$ years. Check: $800 \times 3 = 2\,400$. (c) We use real numbers every day. Achieng's one trip needed whole numbers for counting items and money (3 items, Ksh 135, change Ksh 65), fractions for parts of a whole ($1/2$ kg of sugar) and decimals for prices (Ksh 22.50). These are different ways of writing rational numbers, and rational numbers together with irrational numbers such as $\sqrt{2}$ and π make up the real numbers. Reciprocals are important because dividing by n is the same as multiplying by $1/n$. This turns division problems into multiplication, as in unit price ($90 \times 1/6 = \text{Ksh } 15$), speed and time ($480 \times 1/60 = 8$ hours) and Sacco interest ($2\,400 \times 1/800 = 3$ years). So real numbers and reciprocals together describe everyday life.)

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Number types and classification	Correctly names whole numbers, fractions, decimals, rational and irrational numbers, and classifies the duka numbers as odd, even, prime or composite using factors, with correct handling of 1 and 2.	Names most number types and classifies most duka numbers correctly, with some factor reasoning.	Confuses number types or classifies numbers without using factors.
Evidence for rational and irrational numbers	Uses the decimal conversions and the squaring checks (1.999396, 1.99996164, 2.002225) to explain clearly why $\sqrt{2}$ is irrational and places numbers on the real-number map.	Uses some evidence and states that $\sqrt{2}$ never ends or repeats, but the link to the squaring checks is partial.	States that $\sqrt{2}$ is irrational without evidence, or confuses recurring and non-recurring decimals.

Reciprocals and tools	Defines the reciprocal, finds correct values, checks with $n \times 1/n = 1$, links to unit price, and explains why tables and the calculator differ.	Defines the reciprocal and finds most values correctly, with limited explanation of tools or accuracy.	Confuses reciprocal with another idea or makes calculation errors with no check.
Application and accuracy of calculation	All calculations are correct and shown (135, 65, 15, 8 h, 6 h, Ksh 800, Ksh 400, 3 years) using reciprocals.	Most calculations are correct with some working shown.	Several calculation errors or no working shown.
Answer to the Driving Question	Gives a connected answer tying Achieng's trip to real numbers and to reciprocals turning division into multiplication, using examples from several lessons.	Answers the question using the duka trip but gives few links to other lessons.	Gives a list of facts without answering the question.

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